# CS-104 Retrieval Quiz (ANSWER KEY)

Fall 2026 · Week 6, Friday

Name: _____

*Closed book, no devices. For questions 1–3, circle the one best answer. For question 4, write your code in the space provided (no algorithm, documentation, or test cases needed).*

**1.**

What does this code print?

```
def half(x):
    return x / 2

# --- main program ---
total = half(10) + half(4)
print(total)
```

a) `7`

**b)** `7.0` **[correct]**

c) `14.0`

d) `None`

**2.**

What is wrong with this function?

```
def fahrenheit(celsius)
    return celsius * 9 / 5 + 32
```

a) It should use `print` instead of `return`

b) `celsius` must be assigned a value before the function is defined

c) Nothing; it is correct

**d) The `def` line is missing a colon (`:`) at the end** **[correct]**

**3.**

Assume a function `cube(x)` has been defined that returns `x` cubed. What does `print(cube(1 + 2 + 3))` print?

a) `36`

b) `6`

**c)** `216` **[correct]**

d) `18`

**4.**

A cylinder's volume is pi × radius² × height. Write a function `cylinder_volume(radius, height)` that returns the volume, using `math.pi` (assume `import math` was already done). Then call it for a radius of 2 and a height of 10, store the result in a variable, and print it rounded to 1 decimal place.

**Sample solution:**

```
def cylinder_volume(radius, height):
    return math.pi * radius ** 2 * height

v = cylinder_volume(2, 10)
print(round(v, 1))
```